# DEPARTMENT OF CIVIL ENGINEERING

VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY
BURLA - 768018, ODISHA

# E-NEWSLETTER

2023-2024

## FROM THE HoD's DESK

Dear all,

I am delighted to present to you the most comprehensive edition of our departmental newsletter, spanning from September to April 2023-24. This issue celebrates the remarkable accomplishments and ongoing endeavors within the Civil Engineering Department at VSSUT, Burla. From cuttingedge research initiatives to student triumphs and faculty honors, this newsletter embodies our collective pursuit of excellence in both academicsand innovation. I would like to express my gratitude to our dedicated faculty, enthusiastic students, and committed staff for their valuable contributions. Together, let's continue to drive the future of civil engineering forward.

Thanks
Prof. Sudhanshu Sekhavr Das

# Achievements in Brief

1. Total 137 students got their B-tech degree in civil engineering dept in the 15th convocation during the year 2023.

2. Total 47 students got placed from the Civil Engineering Dept in various companies like Capgemini SE, GenC, Aditya Birla, L&T, Adani, Deloitte, Aess Solutions Ltd, Aquagreen, Iserveu, Reliance Jio Mobility, Pradan, Pregrad, Hindustan Construction Company Ltd, Afcons Infrastructure, Chryso, Acmegrade (2023-24)

3. During the financial years 2023-24, the department has done 8 consultancy projects through various agencies. Under the leadership of our esteemed aculty members as chief consultants, these projects were collectively funded with a total amount of 19,56,405/-

4. There are 26 number of journals published from the department.

5. There are 5 no. of patents granted by various authorities


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# LIST OF PUBLICATION JOURNALS (2023-24)

- Number of SCI Publications: 19
- Number of SCOPUS Publications: 6
- Number of other Publications: 1

1. Pany, R., Rath, A., & Swain, P. C. (2023). Water quality assessment for River Mahanadi of Odisha, India using statistical techniques and artificial neural networks. Journal of Cleaner Production, 417, 137713. Elsevier, (SCIE, IF11.072), https://doi.org/10.1016/j.jclepro.2023.137713.

2. Parhi, S. K., & Patro, S. K. (2024). Application of R-curve, ANCOVA, and RSM techniques on fracture toughness enhancement in PET fiber-reinforced concrete. Construction and Building Materials, 411, 134644. Elsevier, (SCIE, IF-7.4), https://doi.org/10.1016/j.conbuildmat.2023.134644.

3. Mishra, J., Nanda, B., Patro, S. K., & Krishna, R. S. (2024). A comprehensive review on compressive strength and microstructure properties of GGBS-based geopolymer binder systems. Construction and Building Materials, 417, 135242. Elsevier, (SCIE, IF-7.4), https://doi.org/10.1016/j.conbuildmat.2024.135242.

4. Dwibedy, S., & Panigrahi, S. K. (2023). Factors affecting the structural performance of geopolymer concrete beam composites. Construction and Building Materials, 409, 134129. (SCIE, IF-7.693), https://doi.org/10.1016/j.conbuildmat.2023.134129.

5. Dash, P. K., Parhi, S. K., Patro, S. K., & Panigrahi, R. (2023). Efficient machine learning algorithm with enhanced cat swarm optimization for prediction of compressive strength of GGBS-based geopolymer concrete at elevated temperature. Construction and Building Materials, 400, 132814. Elsevier, (SCIE, IF-7.4), https://doi.org/10.1016/j.conbuildmat.2023.132814.

6. Chanda, S. S., Patel, S. K., Nayak, A. N., & Mohanty, C. R. (2023). Performance evaluation on bond, durability, microstructure, cost effectiveness and environmental impacts of fly ash cenosphere based structural lightweight concrete. Construction and Building Materials, 397, 132429. Elsevier, (SCIE, IF-7.4), https://doi.org/10.1016/j.conbuildmat.2023.132429.

7. Parhi, S. K., Dwibedy, S., & Panigrahi, S. K. (2024). AI-driven critical parameter optimization of sustainable selfcompacting geopolymer concrete. Journal of Building Engineering, 108923. Elsevier, (SCIE, IF-7.144), https://doi.org/10.1016/j.jobe.2024.108923.

8. Ijmulwar, S. S., & Patro, S. K. (2024, March). Seismic design of reinforced concrete buildings equipped with viscous dampers using simplified performance-based approach. In Structures (Vol. 61, p. 106020). Elsevier. Elsevier, (SCIE, IF-7.14), https://doi.org/10.1016/j.istruc.2024.106020.

9. Dash, P. K., Parhi, S. K., Patro, S. K., & Panigrahi, R. (2023). Influence of chemical constituents of binder and activator in predicting compressive strength of fly ash-based geopolymer concrete using firefly-optimized hybrid ensemble machine learning model. Materials Today Communications, 37, 107485. Elsevier, (SCIE, IF-3.8), https://doi.org/10.1016/j.mtcomm.2023.107485

10. Pradhan, J., Panda, S., Parhi, S. K., Pradhan, P., & Panigrahi, S. K. (2024). GGBFS-based self-compacting geopolymer concrete with optimized mix parameters established on fresh, mechanical, and durability characteristics. Journal of Materials in Civil Engineering, 36(2), 04023578. (SCIE, IF-3.651), https://doi.org/10.1061/JMCEE7.MTENG-16669.

11. Pradhan, J., Panda, S., Dwibedy, S., Pradhan, P., & Panigrahi, S. K. (2024). Production of durable high-strength self-compacting geopolymer concrete with GGBFS as a precursor. Journal of Material Cycles and Waste Management, 26(1), 529-551. Springer, (SCIE, IF-3.579), https://doi.org/10.1007/s10163-023-01851-0.

12. Prusty, S. R., Panigrahi, R., & Jena, S. (2024). Characterisation and life-cycle assessment of alkali-activated concrete using industrial wastes. International Journal of Environmental Science and Technology, 21(3), 2923-2938. (Springer, SCIE, IF-3.2), https://doi.org/10.1007/s13762-023-05100-7.

13. Panda, S., Pradhan, M., & Panigrahi, S. K. (2024). Optimal Red Mud Replacement Level Evaluation in Conventional Concrete Based on Fresh and Hardened Concrete Characteristics. Journal of Sustainable Metallurgy, 1-16. Springer, (ESCI, IF-2.4), https://doi.org/10.1007/s40831-024-00837-y.

14. Subhadarsini, S., Giri, D., & Das, S. S. (2024). Parametric optimization of bentonite-fly-ash composite core for earthen embankment using Taguchi coupled sunflower optimization algorithm. Innovative Infrastructure Solutions, 9(2), 43. Springer, (ESCI, IF-2.4), https://doi.org/10.1007/s41062-023-01312-5.

15. Parhi, S. K., Panda, S., Dwibedy, S., & Panigrahi, S. K. (2024). Metaheuristic optimization of machine learning models for strength prediction of high-performance self-compacting alkali-activated slag concrete. Multiscale and Multidisciplinary Modeling, Experiments and Design, 1-28. Springer, (ESCI, IF-2.2) https://doi.org/10.1007/s41939-023-00349-4.

16. Acharya, P. K., & Patro, S. K. (2024). Strength, Wear-Resistance, Degree of Hydration, Energy and Carbon Performance of Concrete Using Ferrochrome Waste Materials. Iranian Journal of Science and Technology, Transactions of Civil Engineering, 48(1), 353-362. Springer, (SCIE, IF-1.7), https://doi.org/10.1007/s40996-023-01310-8.

17. Sahu, P., Dalai, R., & Sahu, R. L. (2024). Experimental Study, Using Recycled Aggregates in Mix Design of Granular Sub Base (GSB) Layer "Close Graded: Grading II Gradation". Indian Geotechnical Journal, 1-9. Springer, (ESCI, IF1.4), https://doi.org/10.1007/s40098-024-00881-1.

18. Das, S. K., Nayak, M. K., Patro, S. K., & Suda, Y. (2023). Durability Properties of Ambient Cured Geopolymer Mortar Made from Rice Husk Ash–Based Alkali Activator: A Comparative Study with Conventional Alkali Activator. Advances in Civil Engineering Materials, 12(1), 314-328. (ESCI, IF-1.4), https://doi.org/10.1520/ACEM20230069.

19. Munda, J., & Mohanty, S. (2024). State-of-the-Art Review on Strength Performance of Soil Treated with Silica Nanoparticles. Indian Geotechnical Journal, 1-19. Springer, Scopus, (ESCI, IF-1.4), https://doi.org/10.1007/s40098-024-00868-y.

20. Bhuyan, P. K., Meher, J., & Mohanty, L. (2024). Soil erosion estimation using RUSLE: a case study of Bamnidhi sub basin. ISH Journal of Hydraulic Engineering, 1-8. Taylor & Francis, (Scopus, IF-4.5), https://doi.org/10.1080/09715010.2024.2301721.


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21. Parhi, S. K., & Patro, S. K. (2024). Compressive strength prediction of PET fiber-reinforced concrete using Dolphin echolocation optimized decision tree-based machine learning algorithms. Asian Journal of Civil Engineering, 25(1), 977-996. Springer, (Scopus, IF-1.9), https://doi.org/10.1007/s42107-023-00826-8

22. Behera, H. K., Das, S. S., & Giri, D. (2023). Effects of a two-phase mixing process on warm-mix asphalt by factorial analysis. Proceedings of the Institution of Civil Engineers-Construction Materials, 1-19. (Scopus, IF-1.5), https://doi.org/10.1680/jcoma.23.00039

23. Acharya, P.K., & Patro, S.K. (2023). Structural characteristics of reinforced concrete beams made of waste materials from ferrochrome industry. The Indian Concrete Journal, (Scopus, IF-0.44).

24. Parhi, S. K., & Panigrahi, S. K. (2024). Alkali–silica reaction expansion prediction in concrete using hybrid metaheuristic optimized machine learning algorithms. Asian Journal of Civil Engineering, 25(1), Springer, (Scopus), 1091-1113. https://doi.org/10.1007/s42107-023-00799-8

25. Das, S. K., Behera, N., Patro, S. K., Mustakim, S. M., Suda, Y., & Leklou, N. (2024). Effectiveness of rice husk ashderived alkali activator in fresh, mechanical, and microstructure properties of geopolymer mortar at ambient temperature curing. Journal of Sustainable Cement-Based Materials, 13(2), 213-221. Taylor & Francis, (Scopus, IF4.5), https://doi.org/10.1080/21650373.2023.2262465

26. Prusty, S. R., Jena, S. and Panigrahi, R., (2023). Assessment of acid and chloride resistance of ferrochrome slagbased geopolymer concrete. Indian concrete Journal.

# Awards and recognition (Faculty)

1. Brij Mohan Lal Memorial Prize was presented to Tribikaram mohanty , Ankit Kumar , Prasanna Kumar Acharya , Sanjaya Kumar Patro and Purnachandra Saha for the paper titled performance of structural geopolymer concrete utilising Ferrochrome Ash and Fly Ash as source material published in the journal of the institution of Engineers (India ) Volume 103, Series A, Issue 4.

2. The best faculty award of the year 2023 in Associate professor / Professor category was presented to Prof. Sanjaya Kumar Patro for the significant contributions made on teaching , reasearch , consultancy, project , professional activities and participation in administration to the branch.

3. Felicitation certificate was presented to Professor Sanjay Kumar Patro for his exceptional contribution in publishing five or more indexed journal papers in SCI/SCIE/Scopus in the year 2023

# Student's achievement

1. Saswati Das Choudhary, from Batch 2024, was awarded the Visweswaraya Prativa Puraskar 2023 by the Evergreen Forum on Engineer's Day. She secured 3rd prize in the Youth Gandhi Inter College Quiz Competition 2023 at BJB College and emerged as the champion in the senior category of the Kalinga Quest Quiz 2024.

2. Chinmay Kumar Das and B. Priyajogi at ResCon 2024, organized by the EnPower Research Wing at IIT Bombay. Out of over 100 entries from across India, only 40 projects were selected for poster presentations, and just 10 advanced to the final round. Their project, titled "Healing Cracks, Building Future, Nurturing Sustainability", focused on Bacteria-Based Self-Healing Self-Compacting Geo-Polymer Concrete (BSCGC) and earned them an impressive 7th place at the national level.

3. Ankit Kumar Verma' s accomplishments in 2023-24 with gold at NIT Rourkela and the OUTR Engineer's Cup, and silver at IIIT Bhubaneswar and IIM Sambalpur. In addition to football, Ankit holds silver in kho-kho and volleyball.

4. Manoj Kumar Jena achieved AIR 5 in the BSE Nukkad Natak competition held during Investors Awareness Week 2023. He has also showcased his talents in Nukkad Natak, theater, and stand-up comedy at IIT Kharagpur's Spring Fest (2023 and 2024) and at NIT Rourkela's Nitrutsav 2024.

5. Sourav Ghose achieved 1st place in "Debacles of Past" at Vassaunt 2024.

6. Prabhat Kumar Mahapatra was a cadet in the NCC, where he achieved a B-grade in the C certificate exam. He also achieved the 2nd position in strixture competition during STHAPNA-2023.

7. Gyanaranjan Behera was a core member of Pixels and a cadet in the NCC, holding both B and C certificates. He has excelled in various competitions, securing 1st place in monument drawing at the STHAPNA 2023.

vvvvv8. Ashirbad Dash was an active finalist on the reality show 9XM India's Talent Fight Season 4. In Vassaunt 2023, he secured 1st place in the Voice of Vassaunt competition. He was also recognized as the Emerging Artist of the Year 2023 by Kabichandra Kala Kendra. Sambhunath Biswal participated in VRIDDHI organized by NIT Rourkela and achieved 1st place in the quiz competition at Sthapana 2023.


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# PATENT

Apart from publishing the work in refereed journals, faculty and students are also aware of protecting the ideas and prototypes. Faculty is collaborating with other Dept faculty, Industry, and other institutes for the purpose. Most of these are published and granted by various patent authorities, indicating the competency of the concerned faculty in the area. The details are tabulated below

| Sl no. | Title of the patent | Name of Faculty | Indian/Other | Date of Published |
|---|---|---|---|---|
| 1 | Development of fly ash and ground granulated blast furnace slag blended Geoplolymer concrete paver block | Dr. S Jena & Dr. R.K Panigrahi | Indian | 06/06/2024 |
| 2 | Development of Geoplolymer mortar using alternative alkali activator prepared from blended rice hush ash and sugercane bagasse ash | Dr. S Jena & Dr. R.K Panigrah | Indian | 08/04/2024 |
| 3 | Rice Husk Derived Cement-less Alkali Activated Binder System for Structural Application | Dr.S.K. Patro & B.Nanda | Indian | 01/03/202 |
| 4 | Method and composition of rice husk ash derived alkali activator based cemnet free GPC | Dr. S.K. Patro | Indian | 20/10/202 |
| 5 | Evaluation of High Volume Fly ash Concrete for Rigid Pavemengt Overlays | Kajal Swain | Indian | 13/10/2023 |

# Consultancy (from Industry)

During the year 2023-2024, the construction of the Vice Chancellor's office-cum-residence at VSSUT, Burla got consultancy services for the project, with a consultancy fee amounting to ₹1,18,000. The consultancy fee for vetting the design of SMC office gate services was ₹59,000. AIIMS Bhubaneswar provided consultancy services for its project with a fee of ₹2,39,758. RBI, Bhubaneswar provided a consultancy fee of 1,13,280/-. The department also got consultancy fee ₹1,94,558 from NAC and an amount of Rs. 11,02,586 from CPWD, Sambalpur

# Events:

## SEMINARS

### Ultratech:

The UltraTech Cement Program conducted in the Civil Engineering Department at Veer Surendra Sai University of Technology (VSSUT), Burla, is a significant initiative aimed at bridging the gap between academia and industry. This program focuses on equipping students with practical knowledge and skills related to cement technology and construction practices, reflecting UltraTech's commitment to education and innovation in the field of civil engineering. The event was conducted on 5th April, 2024 by Pritish Sen, MBA graduate in Marketing from IISWBM, Kolkata.The primary objectives of the UltraTech Cement Program at VSSUT Burla was to provide students with insights into the latest advancements in cement technology and the role of cement in sustainable construction. The UltraTech Cement Program at VSSUT Burla is a vital initiative that significantly enriches the educational experience of civil engineering students. By combining theoretical knowledge with practical training, this program empowers students to become skilled professionals ready to meet the challenges of the construction industry. The collaboration between UltraTech Cement and VSSUT underscores the importance of industry engagement in higher education, fostering a generation of engineers who are well-equipped to contribute to sustainable development and innovation in civil engineering.


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BIS:

The Bureau of Indian Standards (BIS) conducted an informative seminar at the Department of Civil Engineering on March 12, 2024 at Veer Surendra Sai University of Technology (VSSUT), Burla, focusing on the importance of standardization in civil engineering and construction. Experts from BIS emphasized the role of adhering to national standards to ensure the safety, quality, and efficiency of infrastructure projects.
The seminar covered topics such as advancements in building materials, safety protocols, and sustainable practices, encouraging participants to align their work with BIS norms. This event aimed to equip students and faculty with the knowledge needed to contribute to the nation's infrastructure development through standardized practices.

BIM:
On August 27, 2023, the Civil Engineering Department organized an extensive 9-hour Building Information Modeling (BIM) Awareness Workshop. The event was graced by Mr. Tushar Patnaik, an expert from BIMedge Pvt. Ltd., Bhubaneswar.The workshop aimed to introduce students to the intricacies of BIM. Participants engaged in hands-on activities such as modeling buildings and utilizing Revit software for both substructure and superstructure design.To keep the students engaged, interactive activities like quizzes were conducted at the end of the workshop.

# DEPARTMENTAL PICNIC

The Civil Department faculty recently enjoyed a refreshing picnic, providing a welcome break from the usual academic routine. Set amidst the serene surroundings of Zero point, it was an opportunity for colleagues to unwind, connect, and strengthen bonds in a relaxed environment.

The day was filled with laughter, engaging conversations, and a range of lighthearted activities that fostered camaraderie and teamwork. From a delicious shared meal to moments spent simply enjoying the peaceful setting, the picnic was a memorable and rejuvenating experience for all. It was a perfect reminder of the strong sense of community within our department and left everyone looking forward to more such gatherings in the future.

## A Friendly Cricket Match:

A Friendly Cricket Match was held between final year students and faculties held on April 11, 2023.In anticipation of the upcoming farewell ceremony for the final-year students, an exhilarating friendly Cricket match was organized between the Faculties and the soon-to-be-graduated students.Prof. Prakash Chandra Swain, Dean of the School of Infrastructure and Planning, inaugurated the event. The match featured the energetic faculties of the Civil Engineering Department (Team SIP) and excited students of the final year students (Team FY).
Dr. Rakesh Roshan Dash led Team SIP, while Shohan Sahoo captained Team FY. It was an intense match, with Team SIP emerging victorious. The audience thoroughly enjoyed the spirited match, making it a memorable day of sports and camaraderie.

# STHAPNA

The Civil department organised its branch fest "Sthapna" with the presence of students from all four years of BTech program.Many events were organised and participated in large numbers.

The Blitz Briefing was a dynamic poster presentation competition aimed at showcasing innovative research and ideas across various disciplines. This event encouraged participants to present their work concisely and engagingly, allowing for effective communication of complex concepts to a diverse audience.

The Civilia was an engaging competition designed to test knowledge in the field of civil engineering. It aims to promote awareness and understanding of civil engineering principles among students and professionals.

Sketch Venture was a resounding success, fostering a love for art and history within the community. The event not only highlighted the beauty of local monuments but also encouraged participants to explore their artistic potential. Plans are already underway for the next edition, promising to be even bigger and better.

In Enigma Grid participants expressed enthusiasm about the unique format, noting that it made learning engaging. Many appreciated the challenge and the opportunity to showcase their expertise. Suggestions for future events included adding more interactive elements and extending the competition to include team-based puzzles.

CiviQuest successfully met its objectives of encouraging community engagement and cultural awareness. With positive feedback and strong participation, plans are already underway for future editions of the treasure hunt, aiming to build on this year's success and further enhance the experience for all involved.

The "Mind Clash" debate competition recently took place, showcasing some of the brightest minds in a spirited exchange of ideas. The event featured teams from various schools, each presenting well-researched arguments on a range of topics. The competition was intense, with participants demonstrating exceptional public speaking and critical thinking skills.The debate topics covered included current global issues, ethical dilemmas, and innovative solutions to societal challenges. Judges evaluated the teams based on their argumentation, rebuttal skills, and overall presentation. The winning team impressed the judges with their coherent arguments and ability to effectively counter the opposition.


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“Enigma Unveil” was a resounding success, providing a platform for artists to showcase their talents and engage with a community of like-minded individuals. The event not only highlighted the importance of creativity but also encouraged participants to push the boundaries of their artistic expression.

Civishotz focuses on capturing urban monuments, possibly under the theme “Urban Frames”. Urban photography, often referred to as cityscape or architectural photography, involves capturing the essence of cities through their buildings, streets, and public spaces.  This type of photography emphasizes the interplay of light, shadow, and architectural details to convey the character and atmosphere of urban environments.

The Crime Investigation Department (CID) competition was organized to foster critical thinking, teamwork, and problem-solving skills among participants interested in law enforcement and crime  investigation..

# Book Published:

1. Ranjita Das, Soumyaranjan Panda, and Saubhagya Kumar Panigrahi(2023): Influence of different Curing Conditions on Properties of Self-Compacting Geopolymer Concrete, Springer Proceedings, Springer.

2. Jharana Pradhan, Soumyaranjan Panda, and Saubhagya Kumar Panigrahi(2023): Impact of alkali activator types on fresh properties of GGBFS-based Self-Compacting Geopolymer Concrete under ambient temperature, Springer Proceedings, Springer.

3. Deepak RanjanMohapatra, Soumyaranjan Panda, RaghvendraSahu, and Saubhagya Kumar Panigrahi(2023): Behavior of ferrochrome sand in the fresh characteristics of GGBFS-fly ash-based self-compacting geopolymer concrete, Springer Proceedings, Springer.

4. Pradhan, J.; Panda, S.; Mandal, R. K.; Panigrahi, S. K. (2023): Influence of GGBFS-based blended precursor on fresh properties of self-compacting geopolymer concrete under ambient temperature.

5. Das, R.; Panda, S.; Sahoo, A. S.; Panigrahi, S. K. (2023): Effect of superplasticizer types and dosage on the flow characteristics of GGBFS based self-compacting geopolymer concrete.Materials Today:

6. Munda, J., Mohanty, S. (2023). State-Of-The-Art Review on Improvement of Strength Characteristics of Soil Using Nano Silica. In: Muthukkumaran, K., Sathiyamoorthy, R., Moghal, A.A.B., Jeyapriya, S.P. (eds) Ground Improvement Techniques. IGC 2021. Lecture Notes in Civil Engineering, vol 297.

7. B. Nanda (2023),"Basic Civil Engineering." Evincepub Publishing, Bilaspur, India. ISBN: 9789356735743 .

8.Nayak, A., Kar, A. K., (2024). Study and prediction of built up land use for Bhubaneswar and its corresponding effects. In: Sethi, S., Mahmud, M., Pradhan, S. K., Sethi, R. (eds) Industry 4.0.


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# Farewell:

## Students farewell:

The farewell event of the department of civil engineering - Amantrya was organised on 13th of April to bid farewell to the outgoing batch of 2023. The event was held successfully with the presence of HOD , civil engineering Dr. Rakesh Roshan Dash , dean of students welfare Prof. Sanjay Kumar Patro , Dean of school of infrastructure and planning Prof. Prakash Chandra Swain , other faculty members and all the four batches of BTech students .Faculty members and students delivered heartfelt speeches and students were felicitated for their academic and extracurricular achievements. A video montage capturing memorable moments from the student's academic years was presented invoking nostalgia. Dance, music and drama performances were presented to enrich the cultural evening and to leave a lasting impression on the graduating students.

## Teachers farewell:

The Civil Engineering Department recently gathered to bid farewell to honourable Prof Amarnath Nayak, marking the end of an era. The ceremony was a heartfelt tribute to his dedication, passion, and expertise. The Civil Engineering Department extends heartfelt gratitude and wish him all the best in their future endeavors and look forward to staying connected.

2023 also gave heartfelt gratitude for Prof. PC Swain invaluable guidance, wisdom, and dedication. His passion for knowledge has not only inspired us but has also shaped our academic journeys profoundly. His impact will undoubtedly resonate here for years to come, and while we bid him farewell, we know that his legacy remains in each one of us.

As Dr. B. Nanda embark on this new journey, we want to thank him for the years of inspiration, mentorship, and support. His dedicationhas left a lasting mark on us all, and his wisdom will continue to guide us. Wishing him happiness, health, and endless success in this next chapter.

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